Abstract

Large language models are increasingly used in scientific workflows, but their practical value is limited by unsupported certainty, weak handling of evidence conditions, and poor retention of prior corrections. We present Mirror Reflection Agent, a reliability-oriented architecture that separates candidate cognition from final output and inserts an explicit mirror-critique stage between them. The system constructs a structured intermediate state, critiques it with typed verdicts such as revise, retrieve, wait, diverge, and pass, and stores correction lineage and reusable strategy signals in memory. Unlike generic self-refinement pipelines, the proposed architecture externalizes intermediate cognition and makes revision control auditable. We instantiate the design with scientific and commonsense knowledge layers, a deterministic critique module, project-local memory, and replayable run audits. We argue that the key contribution is not a claim about consciousness or subjective awareness, but a practical control mechanism for safer reasoning. The paper positions the architecture as a reliability method for AI for Science, especially in tasks where evidence provenance, uncertainty, and abstention matter as much as answer accuracy. We outline an experimental program that can be executed by a solo researcher using a small benchmark suite, lightweight open models, and API-based baselines, focusing on hallucination reduction, revision quality, correction retention, and bounded abstention.

Introduction

Scientific use of large language models requires more than fluent answers. A useful system must distinguish strong evidence from weak prior knowledge, preserve source conditions, reduce unsupported certainty, and remember prior corrections without blindly repeating them. Existing agent pipelines often improve outputs by adding reflection or retrieval, but they still leave three recurring problems:

- intermediate reasoning is not represented as an inspectable object
- critique outcomes are not encoded as explicit control decisions
- prior corrections are not preserved as structured reusable lineage

This paper addresses those gaps with Mirror Reflection Agent, a structured reliability-control architecture. Rather than directly producing a final answer, the system builds a candidate cognitive state, critiques it with a mirror layer, and then either revises, retrieves memory, diverges to a new scaffold, defers, or finalizes. The design is engineering-oriented and auditable. It does not claim subjective awareness, consciousness, or human-like introspection.

Our central claim is narrow and testable: explicit cognitive state plus typed critique plus correction-lineage memory can improve reliability in scientific and science-adjacent reasoning tasks.

Core Contribution

The deepest contribution of the architecture is not generic self-reflection. It is the conversion of reliability problems into explicit state and explicit control flow.

The architecture contributes three linked mechanisms:

1. \textbf{State externalization.} Candidate reasoning is represented as a structured \texttt{MindState} with claim, evidence, assumptions, alternatives, confidence, and risk.
2. \textbf{Verdict-typed oversight.} Critique is represented as a structured \texttt{MirrorVerdict} with explicit actions: \texttt{pass}, \texttt{revise}, \texttt{retrieve}, \texttt{wait}, and \texttt{diverge}.
3. \textbf{Correction-lineage memory.} Memory stores not only facts, but also prior bias alerts, correction actions, and reusable strategies.

This makes the system materially different from prompt-only reflection. The mirror is not just another text prompt. It is a decision layer that governs whether the system should answer, revise, retrieve, or abstain.

Related Work

This paper sits at the intersection of self-refinement, reflective agents, retrieval-augmented generation, and reliability-oriented AI for Science systems. Prior work such as Self-Refine and Reflexion showed that iterative self-feedback can improve outputs, while CRITIC and Self-RAG demonstrated the value of critique and retrieval in controlled generation. The present work differs in emphasis. It does not primarily contribute a new prompting trick or a new retrieval policy. Instead, it contributes an explicit reliability-control structure in which intermediate cognition is externalized, critique outcomes are typed, and correction lineage is preserved as reusable memory. In the scientific-reasoning setting, this distinction matters because many failures are not pure factual hallucinations; they are failures of scope, condition handling, evidence authority, and repeated overclaiming. The paper therefore frames Mirror Reflection Agent as an auditable control architecture rather than as a purely generative enhancement.

Architecture

The system follows:

```
input -> knowledge layers -> cognition -> mirror critique -> revise/retrieve/diverge/wait/pass -> output
-> memory write-back
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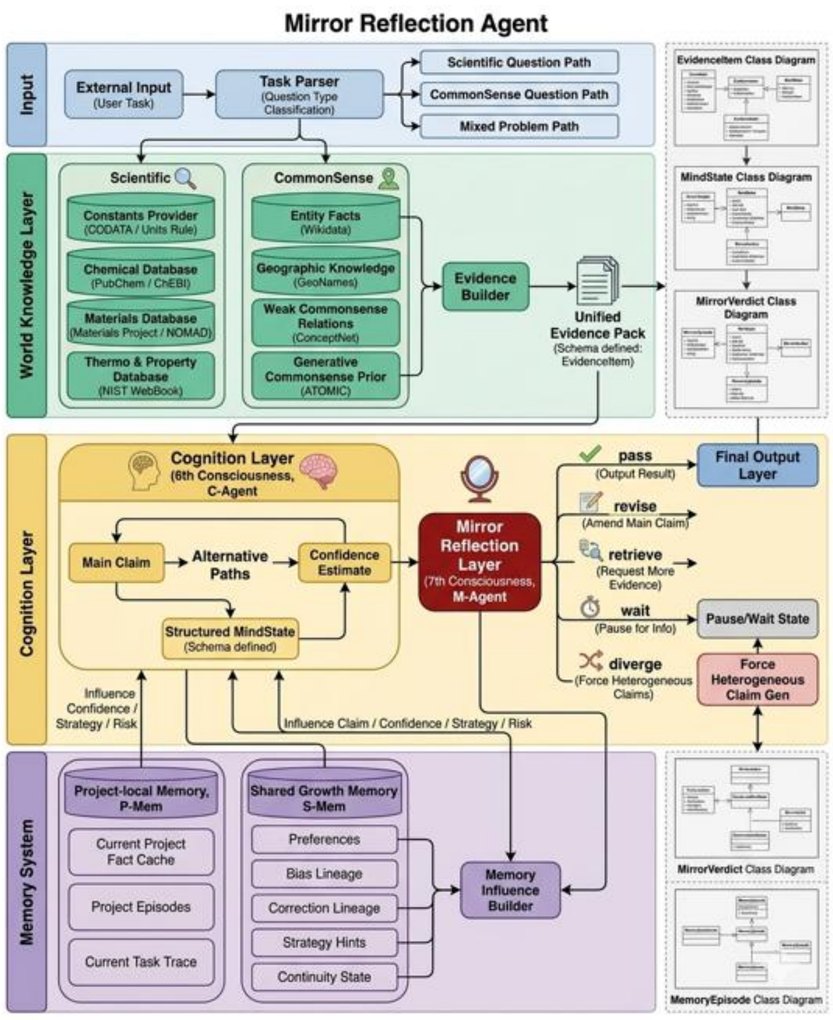

Overall engineering schematic of Mirror Reflection Agent. The figure shows the relation between the scientific knowledge layer, commonsense knowledge layer, cognition layer, mirror reflection layer, and memory system. This should remain the main system overview figure because it clearly exposes the audit path from input to final output.

Scientific Knowledge Layer

The scientific layer is responsible for higher-confidence, condition-sensitive evidence. Its role is to parse scientific entity/property questions, resolve entities and normalize units, route across structured scientific sources, and build evidence packs with provenance and warnings. This layer is the source of hard constraints.

Commonsense Knowledge Layer

The commonsense layer is responsible for softer background priors. Its role is to resolve general entities and relations, provide context when prompts rely on implicit world knowledge, and distinguish structured facts from weak commonsense priors. This layer should not be treated as equivalent to scientific evidence. Its proper role is contextual support, not unrestricted claim justification.

Cognition Layer

The cognition layer builds a structured MindState containing current input, task goal, main claim, evidence, hidden assumptions, alternative paths, confidence, self-risk, proposed action, and self-state. This step externalizes an otherwise opaque intermediate draft.

Mirror Layer

The mirror layer reviews the MindState and returns a MirrorVerdict. Current implemented checks include premature convergence, concept blending, evidence gap, overclaiming, weak commonsense presented as fact, structured-fact priority violation, unit inconsistency, condition-scope violations, stale template reuse, and divergence need. The mirror is best understood as a reliability governor.

Memory Layer

The memory layer contains project-local memory and optional shared-growth memory. The key memory signals are fact hints, context hints, bias alerts, correction hints, strategy hints, and divergence triggers.
This gives the architecture a notion of bounded continuity across runs.

Methods

The current implementation is deterministic in its control structure. A run begins by building memory influence from prior episodes, then constructing scientific and commonsense evidence packs, then generating a candidate MindState. The mirror layer then reviews that state and emits one of five verdicts: pass, revise, retrieve, wait, or diverge. If the verdict is pass, the system returns a bounded output and writes a memory episode. If the verdict is revise, the cognition layer rewrites the claim under the guidance returned by the mirror. If the verdict is retrieve, memory signals are refreshed and re-applied. If the verdict is diverge, the system is forced to adopt a different explanatory scaffold. If the verdict is wait, the system emits a deferred answer.

This structure is important for evaluation because it allows us to separate multiple reliability phenomena that are usually conflated in end-to-end generations: evidence insufficiency, stale-template reuse, prior correction reuse, and calibrated abstention.

Why This Architecture Matters

The architecture addresses several practical problems in current LLM agents.

Knowledge Types By Epistemic Role

Scientific evidence and commonsense priors should not have equal authority. The architecture makes that distinction explicit and allows the mirror layer to prevent weak priors from being promoted into factual claims.

Abstention As A First-Class Outcome

In scientific settings, refusing or deferring can be safer than answering. The wait verdict supports bounded abstention when evidence is insufficient or inconsistent.

Cumulative Correction

Most systems retrieve facts but do not retrieve prior mistakes as reusable constraints. Here, prior corrections become structured signals that can influence later behavior.

Auditability

Each run produces replayable audit information showing memory influence, cognition state, mirror verdicts, and final output.

Problem Setting

The problem setting targeted in this paper is bounded scientific reasoning under uncertainty. The paper does not ask whether the architecture can discover new laws of nature or replace domain scientists. It asks whether an LLM-based system can be made more reliable in cases where the answer must respect evidence provenance, scope, and correction history. This narrower framing is important both scientifically and experimentally. It aligns the architecture with realistic deployment conditions and makes the claims testable by a single researcher.

What This Architecture Is Not

The paper should explicitly reject several overstatements.

- It is not a consciousness model.
- It is not evidence of subjective awareness.
- It is not an autonomous scientist.
- It is not a general solution to continual learning.

The architecture supports bounded reliability improvement under explicit control. That is already a meaningful contribution.

Current Blind Spots

The present system still has important limitations. The agent can reuse corrections, but it does not autonomously redesign its own rules, objectives, or schema. That is controlled adaptation, not full self-improvement. The memory system preserves structured lessons, but it does not yet maintain rich multi-session dialogue continuity with evolving user preferences, conceptual drift, and conflict resolution.
The mirror works well on explicit risks such as overclaiming or evidence gaps, but may still miss subtle errors in decomposition, framing, or hidden variable choice. The current system is therefore better positioned for reliable scientific QA and bounded explanation than for open-ended hypothesis generation.

Experimental Direction For A Solo Researcher

The experimental plan should be designed for limited compute. The main principle is not to beat frontier models on absolute science ability, but to show that the architecture improves reliability properties under practical resource constraints.

Three-Track Evaluation

Track A: Internal reliability benchmark

- concept blending cases
- premature convergence cases
- correction retention cases
- memory pollution cases
- bounded control cases

Track B: Small scientific QA benchmark

- 100 to 300 manually curated questions
- chemistry, materials, physical constants, unit-sensitive queries
- answers backed by structured sources already close to the current implementation

Track C: Longitudinal correction benchmark

- families of related prompts across sessions
- repeated mistakes with corrected variants
- tests whether prior corrections are retrieved, applied, ignored, or overgeneralized

Feasible Baselines

The following baselines are realistic without major infrastructure:

1. Direct answer baseline
2. Structured cognition without mirror
3. Mirror without memory
4. Mirror plus local memory
5. Mirror plus dual-layer memory

Models And Compute

A solo-researcher-friendly setup could use local 7B to 14B instruction-tuned open models for repeatable runs, rented single-GPU cloud instances only for batch experiments, and API models only for a small comparison slice, not the entire study.

Metrics

Primary metrics include answer correctness, hallucination rate, concept-mixing rate, unsupported-certainty rate, revision success rate, correction retention rate, abstention quality, unit-consistency rate, and condition-scope preservation rate. Secondary metrics include average number of revisions, average confidence, retrieval hit rate, and memory pollution risk.

Evaluation Logic

The core evaluation logic is architecture-first. The strongest comparison is not between different foundation models, but between the same underlying model wrapped in different control structures. This yields a clean ablation argument: if reliability improves while the underlying model remains fixed, the improvement is attributable to the architecture rather than to raw model scale.

Experimental Results From The Current Implementation

We ran the existing project benchmark directly from the repository without changing code:

```
python3 -m evals.run\_evals --format json
```

This benchmark currently contains four cases and compares local_only against dual_layer. Even though it is still small, it already shows the intended architectural effect.

Metric Summary

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\begin{table[H]
\centering
\begin{tabular{lccc
\toprule
Metric & local_only & dual_layer & delta \\
\midrule
Premature convergence rate & 1.000 & 0.000 & -1.000 \\
Revision success rate & 0.500 & 1.000 & +0.500 \\
Correction retention rate & 0.000 & 1.000 & +1.000 \\
Shared-memory pollution risk & 0.000 & 0.000 & 0.000 \\
Average confidence & 0.6075 & 0.5625 & -0.0450 \\
\bottomrule
\end{tabular
\caption{Current benchmark summary from the existing implementation.
\label{tab:current-results
\end{table
```

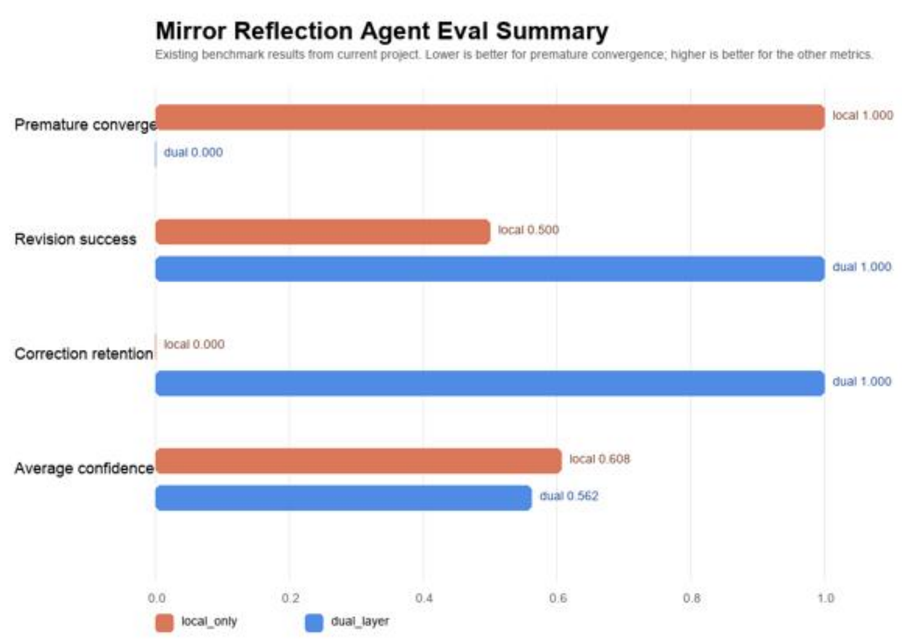

Summary plot of the current benchmark. The dual-layer setting eliminates premature convergence in the current suite and substantially improves revision success and correction retention, while slightly reducing average confidence.

Analysis Of Figure 2

Figure \ref{fig:benchmark shows the current logic of the paper clearly. The main gain is not that dual-layer memory makes the system more assertive. It makes the system more controlled. Premature convergence drops from 1.000 to 0.000, which means the architecture is less likely to lock into an overconfident conclusion too early. Revision success rises from 0.500 to 1.000, indicating that the mirror-plus-memory loop is not just triggering revisions, but helping those revisions land in a safer final state.
Correction retention rises from 0.000 to 1.000, which is especially important for the manuscript because it supports the claim that structured correction lineage can influence later behavior in a measurable way. Average confidence decreases slightly from 0.6075 to 0.5625; in this setting that should be interpreted as improved calibration rather than degradation, because the target behavior is bounded reliability rather than maximal certainty.

Case-Level Interpretation

In concept_blending_shared_recovery, dual-layer memory retained the target correction while local-only did not. In premature_convergence_shared_bias_alert, dual-layer memory removed premature convergence and preserved the shared correction signal. In project_local_precedence, the system preserved project-local strategy priority while still using shared memory as advisory context. In bounded_control_case, dual-layer memory did not degrade already-good behavior.

What These Results Demonstrate

These results do not prove broad scientific superiority. They demonstrate a narrower and more credible point: the architecture improves reliability-oriented behavior under correction and memory. This is the right level of claim for the current paper.

Discussion

The current benchmark is intentionally small, but it already reveals the kind of gain that matters for this architecture. The system becomes less prone to premature commitment, more likely to preserve prior corrections, and more willing to trade raw confidence for bounded behavior. For a scientific assistant, that trade is desirable. A slightly less confident answer that respects evidence limits is often more valuable than a fluent but overgeneralized one.

The results also suggest that the architecture is especially well suited to settings where the central risk is not random error, but recurrent patterned error. In those cases, correction-lineage memory and verdict-typed control give the system a mechanism for avoiding the same mistake under a new wording. That is a more realistic notion of practical improvement than simply raising answer accuracy on a static one-shot benchmark.

Journal Strategy

The recommended primary target is Machine Learning: Science and Technology, because its scope explicitly includes methodological advances in machine learning motivated by scientific problems. Strong fallback venues include Frontiers in Artificial Intelligence, with npj Artificial Intelligence or Communications AI & Computing as more selective stretch targets if the next experiment round becomes substantially stronger.

Limitations

The current prototype does not yet demonstrate strong continual learning, robust multi-session conversational identity, broad scientific discovery ability, or comprehensive critique coverage for all failure modes. These limits should remain explicit in the manuscript.

Conclusion

Mirror Reflection Agent is best understood as a structured reliability-control architecture rather than a claim about machine consciousness or autonomous scientific discovery. Its value lies in externalizing intermediate cognition, enforcing typed critique decisions, and preserving correction lineage across runs. The current implementation already supports a credible reliability narrative: less premature convergence, stronger revision success, and better correction retention under bounded confidence. That is a realistic and publishable foundation for the next iteration of experiments.